UBICOMP ISWC2025

14-16 Oct. 2025
Helsinki, Finland

Globe viewer and POI service

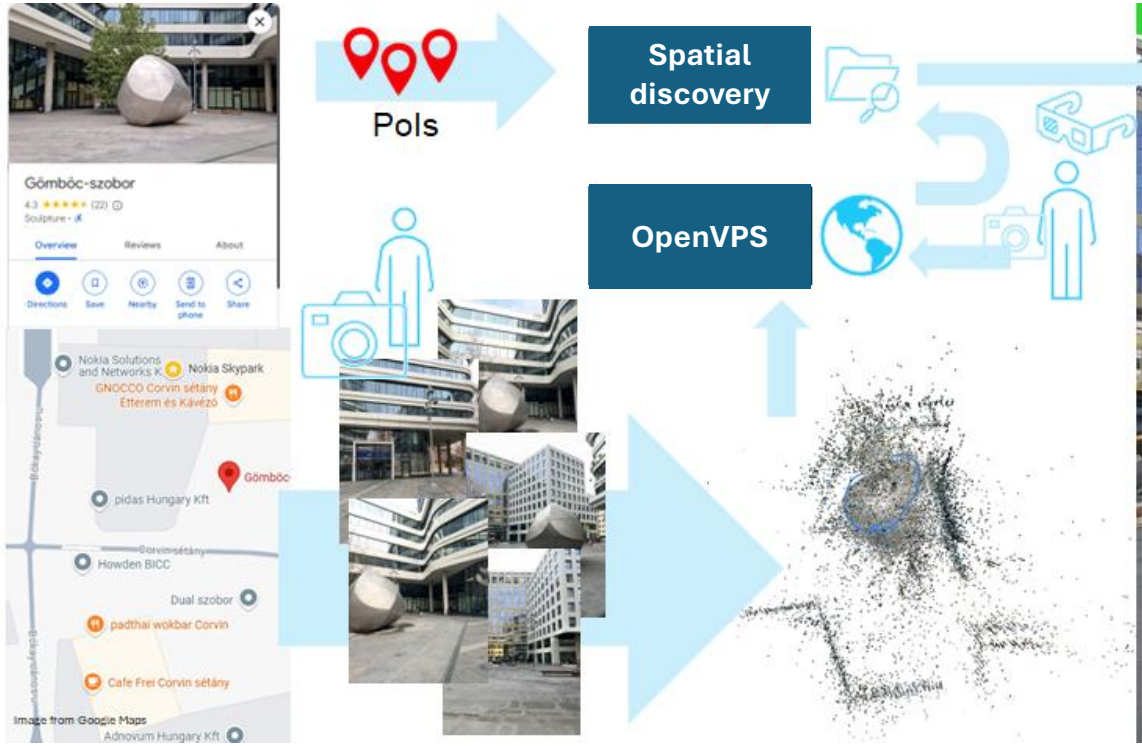
Dr. Gábor Sörös

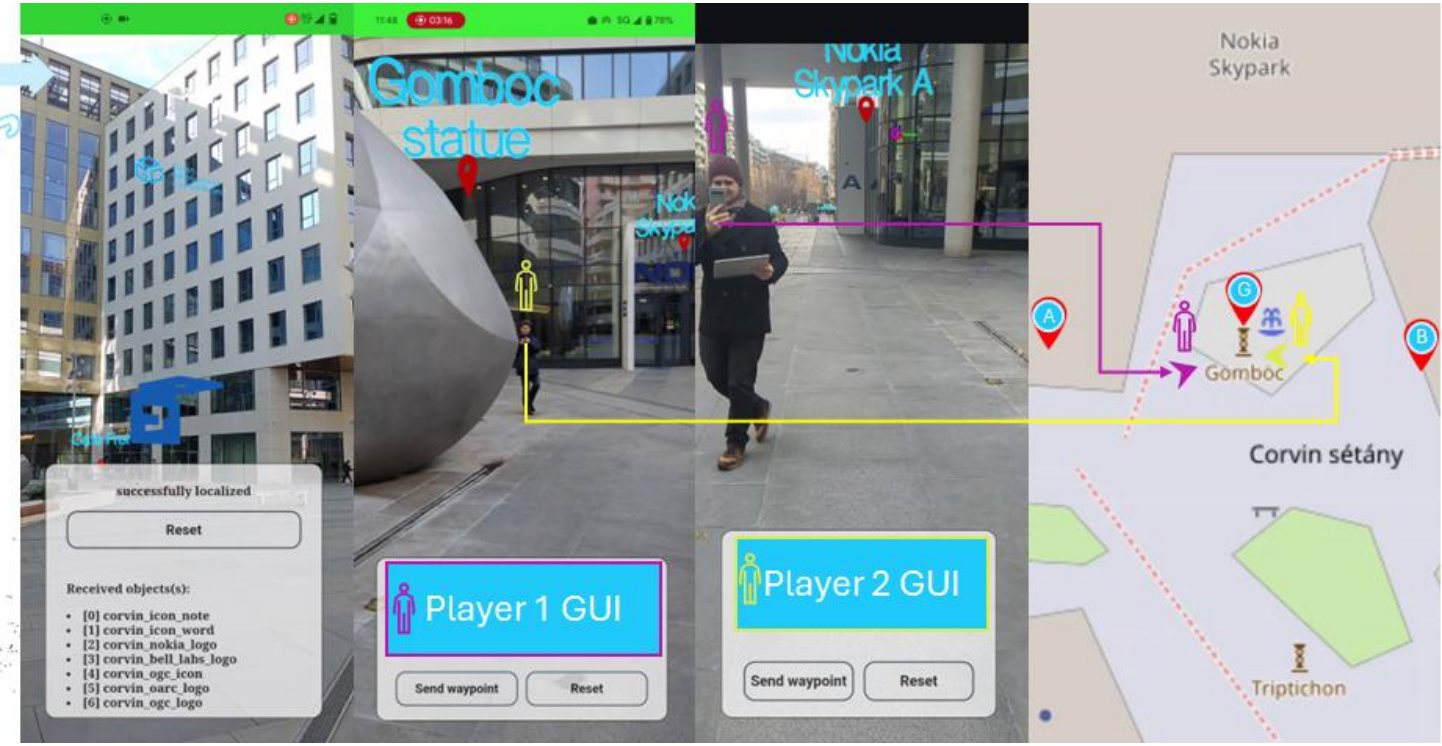
Open AR Cloud & Nokia Bell Labs

OPEN AR CLOUD

OpenVPS + Point of Interest service

Finding nearby restaurants, shops, etc.





Point of Interest (POI) service



Author:
Balázs Ladjánszki

Service that collects data about nearby features based on a text query

The API takes latitude, longitude, and a keyword

We wrapped Google Maps APIs and OpenStreetMap APIs (relay)

The two data providers complement each other, they may find places the other cannot. We merge the duplicates.

All information is returned in OGC POI format (draft standard of Open Geospatial Consortium)

```
"type": "Feature",
"geometry": {
  "type": "Point",
  "coordinates": {47.48601, 19.07941} // at this real-world location
},
"properties": {
    "featureID": 693842,
    "name": {"name" : "Gomboc statue"}, // this gets rendered in the AR view
    "hasFeatureOfInterest": {
        "href": "https://en.wikipedia.org/wiki/G%C3%B6mb%C3%B6c"
    },
    "category": {
      "category": "theatre", "categorySystem": "osm"
    }
          ...
  }
```

OGC POI standard (draft)

Try it!

https://Your URL/locations?lat=60.185484&lng=24.827222&textQuery=restaurant

Location-based augmented reality demo

In-situ visualization of Point of Interest data

- Browser app with WebXR (always-on future with smartglasses)
- Visual positioning → GeoPose (WGS84 lat, lon, h, ENU orientation)
- Content retrieval around GeoPose
- Alignment of AR session with the world
- Simple WebGL rendering engine
- Supported contents: primitives, GLTF model, text, OGC PoI
- Points of interest downloaded from PoI service and shown in live AR
- Applications: restaurant rating, tourism, navigation, etc.

Try it: https://sparcl.orbit-lab.org/
Select Experiment / SearchByText
(We present this app in the second part of this tutorial)

Try it!

Code of service: https://github.com/OpenArCloud/oscp-poi-service
Code of client: https://github.com/OpenArCloud/sparcl/tree/main/src/experiments/oarc/searchbytext



Full video: https://www.youtube.com/watch?v=7Ua7Tpan2HI

Globe viewer

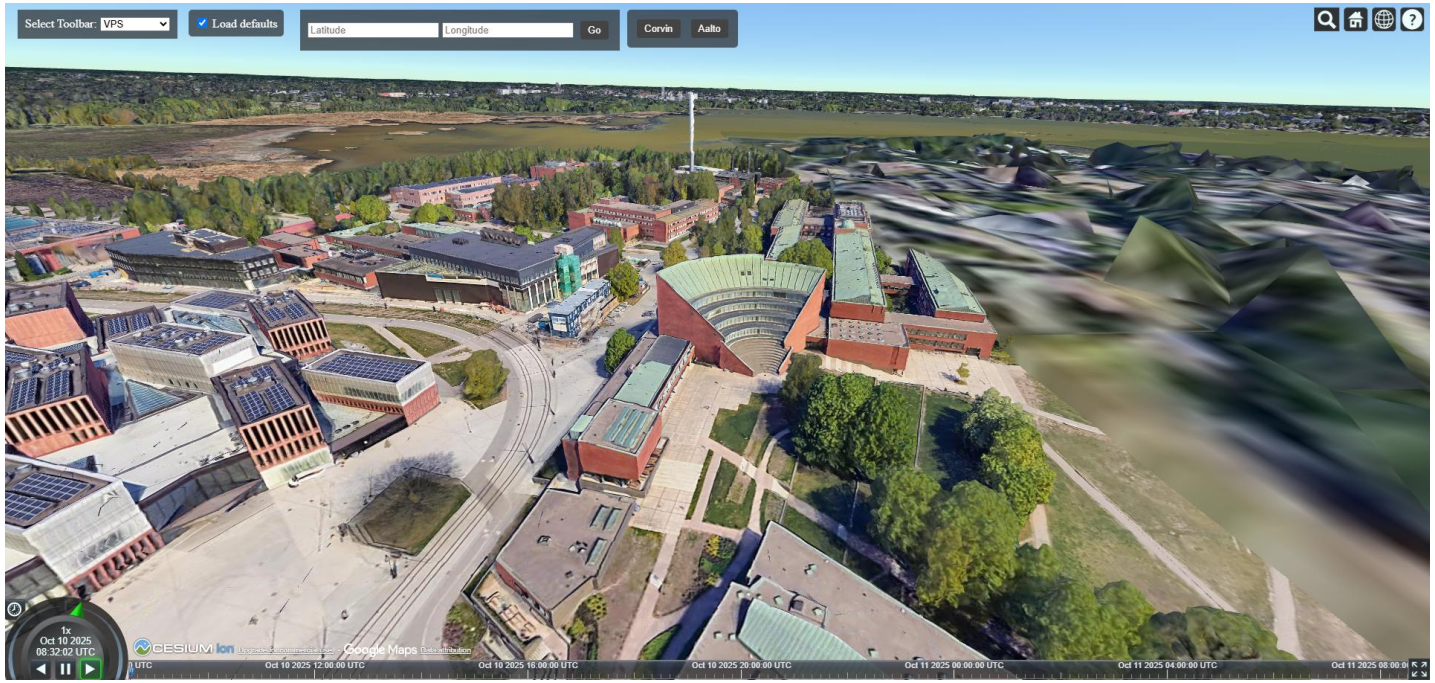


Author:
Balázs Ladjánszki

A Cesium-based globe viewer, using Google Photorealistic 3D tiles

Multiple goals:

- Testing the OpenVPS
- Testing the POI search service
- Debugging location-based AR experiences
- (Authoring 3D contents at geolocations)



https://github.com/OpenArCloud/cesium-viewer

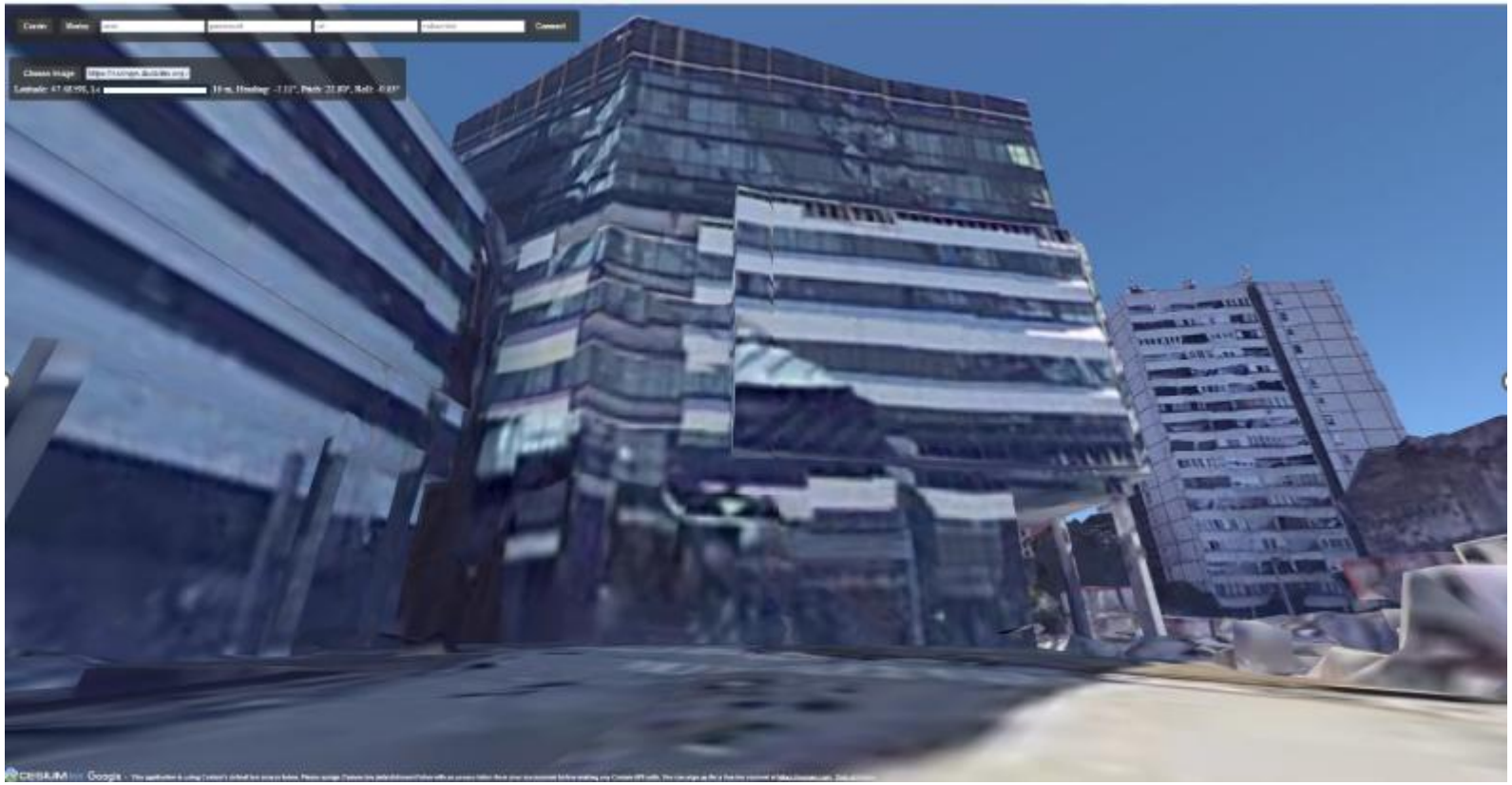
VPS testing
First-person view of the estimated pose

https://oarcvps.duckdns.org:8049/

Try it!



Query image
(JPEG, with GPS coordinates in EXIF)



Globe viewer with camera set at the estimated pose

Video: https://youtu.be/D2AGLcQxROg

POI search testing
Retrieved POIs visualized on the map

https://oarcvps.duckdns.org:8049/

Try it!

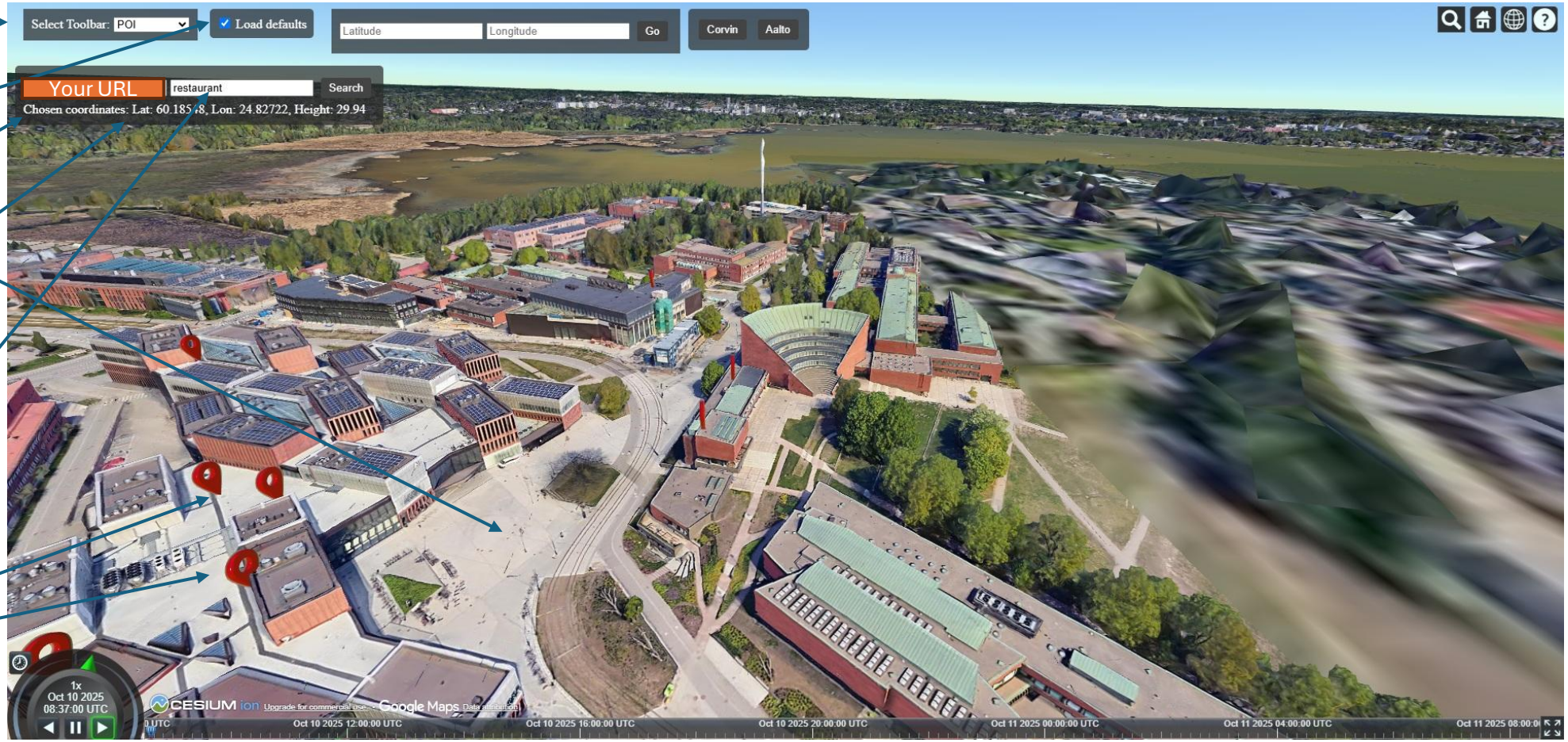
Select POI

Enter server address (or load default - mine)

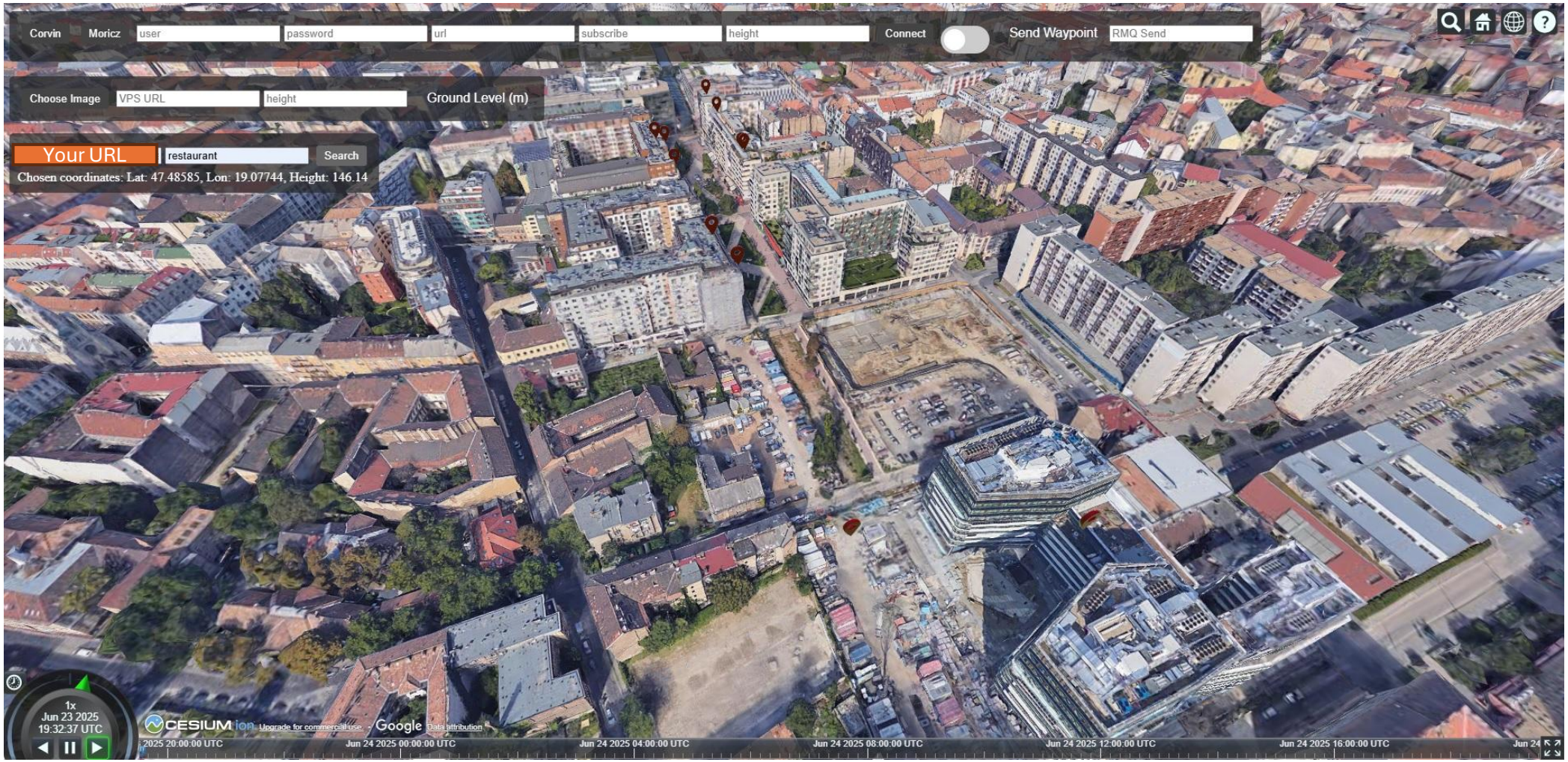
Click on the 3D model to specify search center (200m radius)

Enter search term and press Search

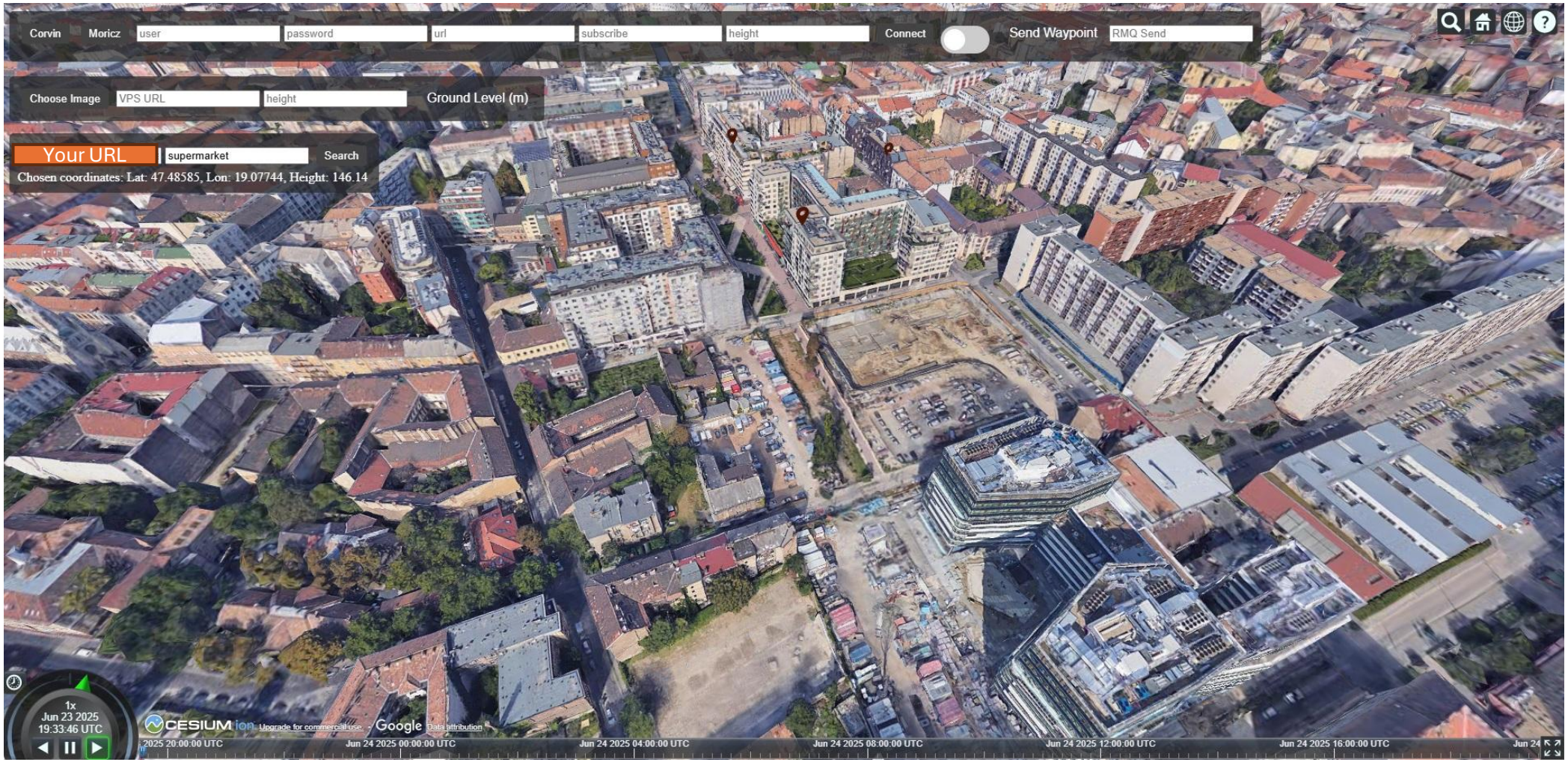
Retrieved POIs



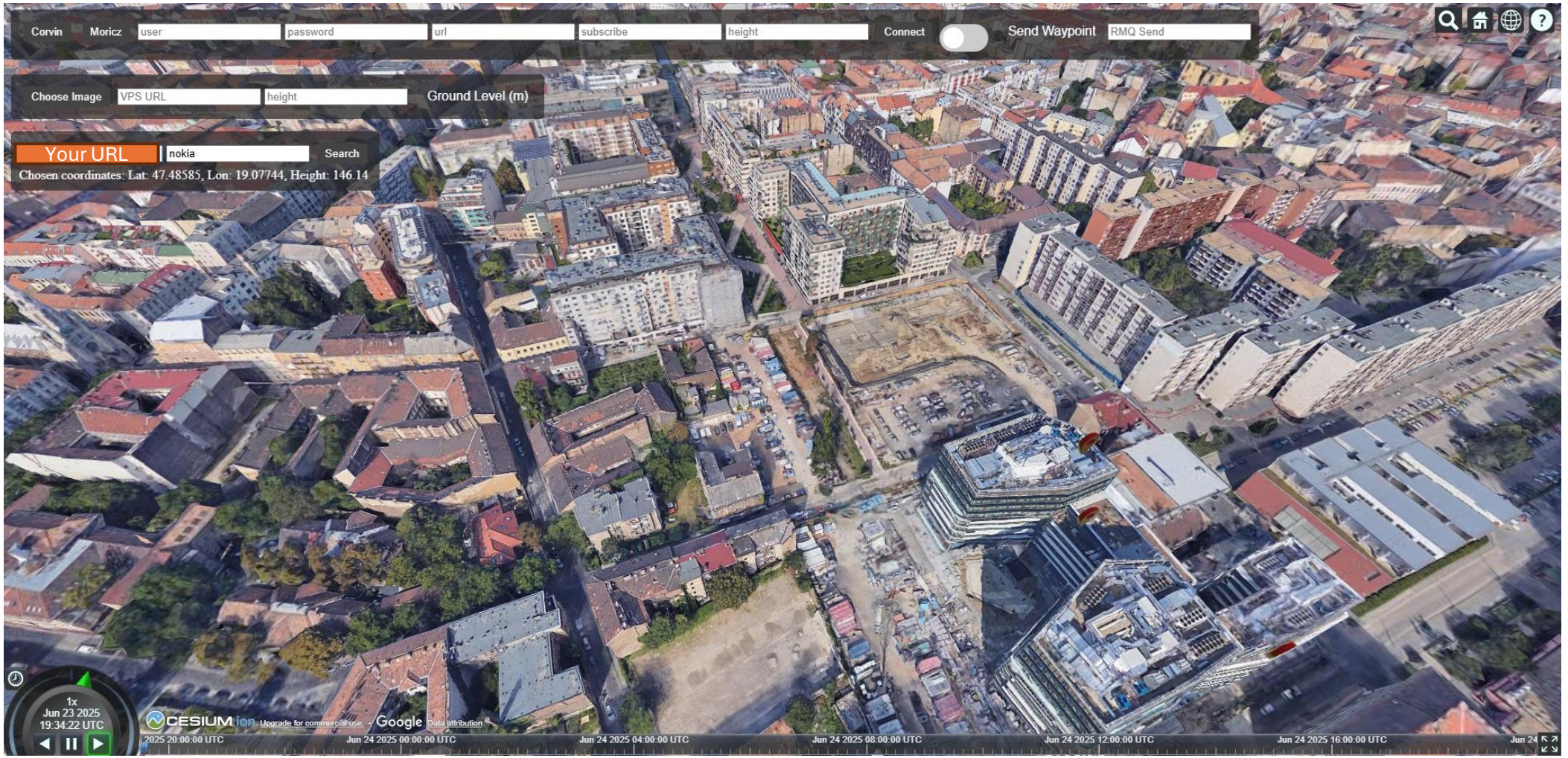
Search: restaurant



Search: supermarket



Search: Nokia



Pose streaming (experimental)

Details later in the tutorial

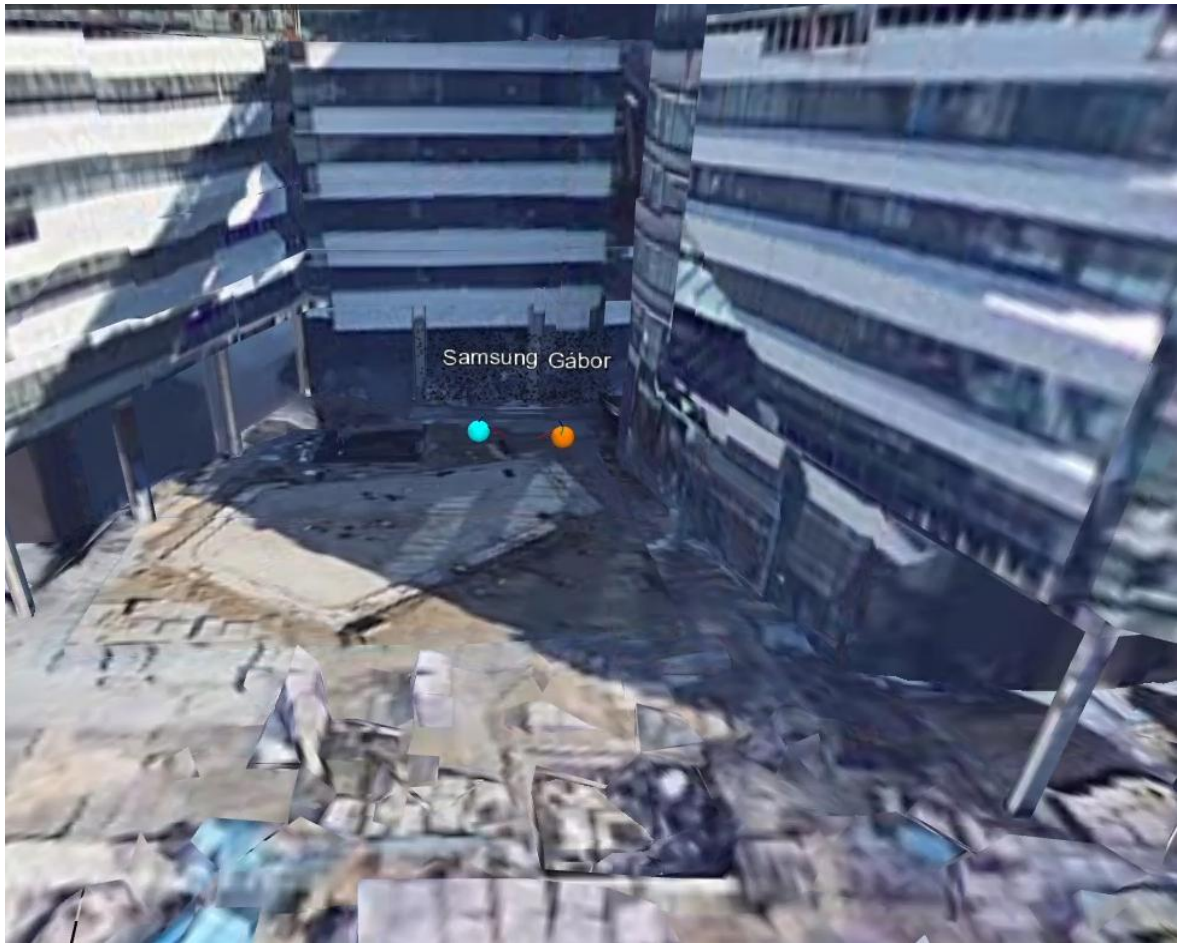


RabbitMQ message broker

AR clients determine own GeoPose
via VPS + local motion tracking

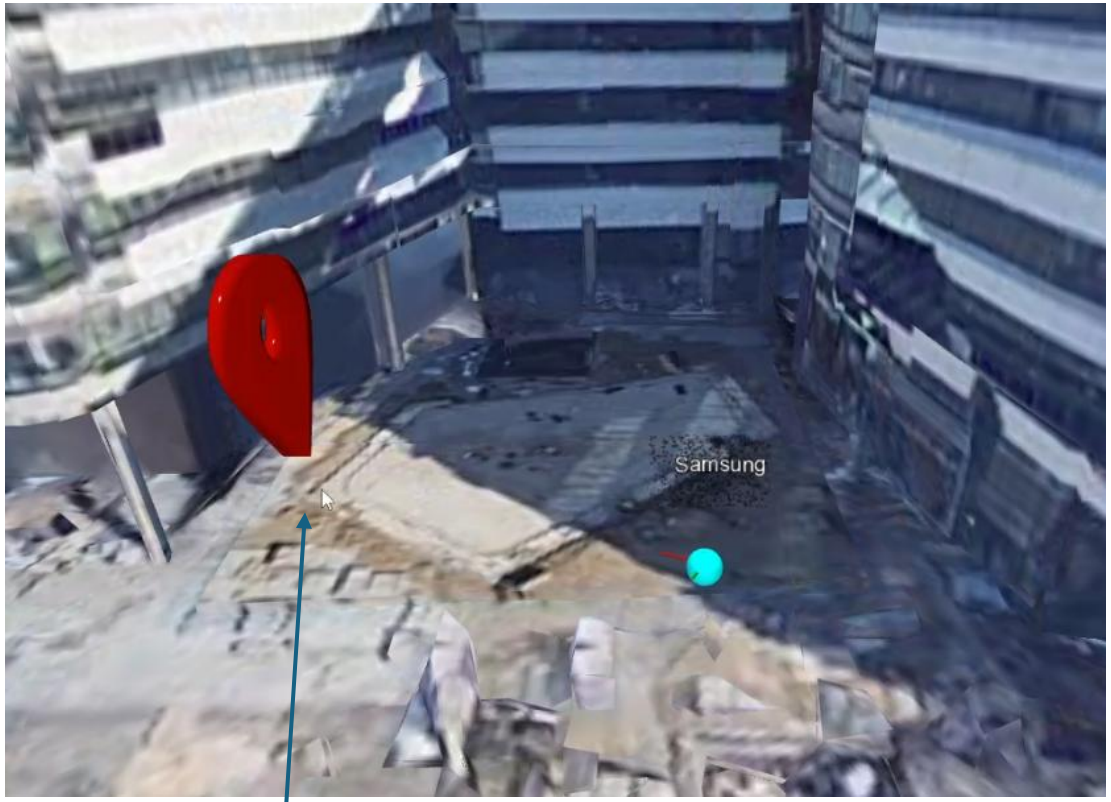
AR clients stream out pose on a shared topic

Globe viewer can visualize the current GeoPoses

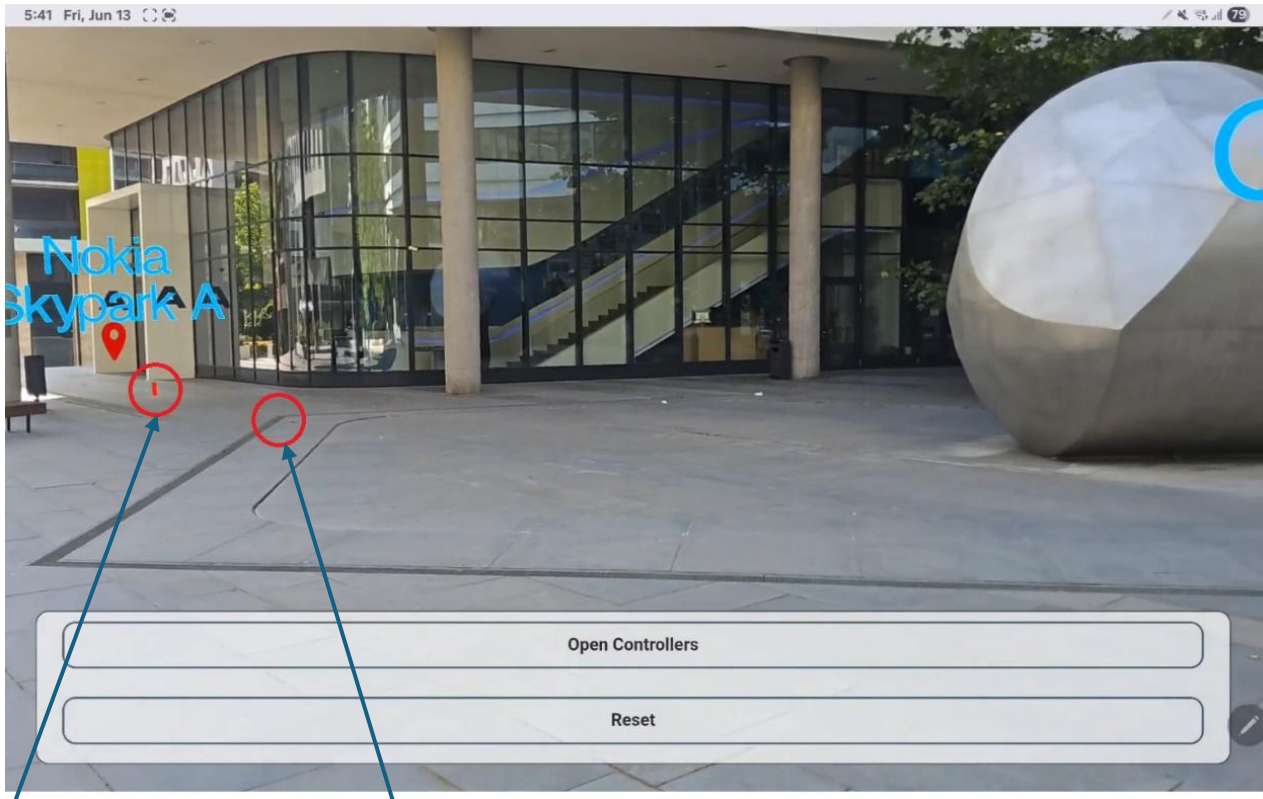


Sending waypoints to AR clients

Globe viewer can send “waypoint” markers to AR clients on a dedicated topic



CLICK



Waypoint in AR

Waypoint should be here